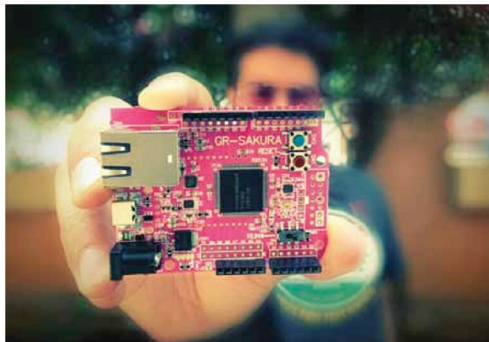


What's New in Open Source Development Boards

Here is a sneak peek at open source development boards with a focus on GR-Sakura—the latest entrant to this segment



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Earlier, an electronic development board was primarily used by engineers working on a prototype or by innovative hobbyists. But today, every aspect of our world has an electronic connection. So it isn't surprising that developer boards have advanced to keep pace with the times.

What's available?

Development boards from the Italian Arduino family of boards are the most famous ones. Because of the open source hardware and software that they feature, these boards are inexpensive and have a bundle of clones to choose from. Their simple coding environment attracts a lot of beginners and non-professionals.

If you want something more than a basic microcontroller-based board, you have BeagleBoard. BeagleBoard

is basically a fanless single-board computer powered by a decent ARM Cortex processor. With low battery life, BeagleBoard is not really apt for most of the applications that the Arduino trumps in due to very low power consumption.

Another major open board is the PandaBoard. It is probably one of the most popular, highly specified development boards available—it comes with a dual-core processor clocked at 1 GHz and is capable of handling 1080p video streaming with ease.

What are the developments?

The release of Raspberry Pi last year took the hacker world by a storm. A single-board computer was available for a mere \$25-35, and was sold out just as quickly as it was made available for purchase online, crashing some e-shop servers in the process.

Raspberry Pi is a strong competition to BeagleBoards and PandaBoards. It comes equipped with 256MB RAM and its own ARM CPU and Broadcom GPU. It also features Ethernet, HDMI and a microSD slot. It is definitely the first choice for those who plan to boot Linux variants and run some software to turn it into a media centre or a gaming system.

A few years ago, NXP too had come out with a development board named 'mbed board.' Priced at \$60, it comes in a 40-pin DIP package and is small enough to fit onto a breadboard easily. The best part about this board is its development environment including the editor, compiler and library. Reference documents are all completely Web-based, freeing the user from having to install or maintain any software on the host system. NXP also helps mbed beginners by providing them a HelloWorld.bin file along with a guide to get started on the board at the product page on mbed.org.

Variety is something that we all want to see in the market. Renesas correctly understood that and came out with a cherry blossom (sakura) flavoured board called 'GR-Sakura.' It is the latest board on the block. At first glance, it looks like a board that has the best of both—the Raspberry Pi and the Arduino families.

Things to note

1. The Sakura board uses a 5V power supply rather than the 9V supply seen on most Arduinos.
2. Although the power supply is 5V, the board actually runs on 3.3V. Therefore input signals should be clamped to 3.3V as wrong voltages can damage your board.